

School of Engineering

Pokhara University

Sub: NLP

Assignment 2

Students are required to write the answer either by handwritten or by typed, and create a single PDF and upload it in the google classroom. Write the question as well.

Chapter 1: Introduction to NLP

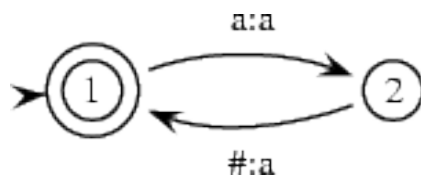
1. Define Natural language processing. Discuss in, brief, the challenges in NLP.
2. Differentiate between Natural Language Understanding (NLU) and Natural Language Generation (NLG).
3. Explain the different phases in NLP.

Chapter 2: Localization and Language computing

4. Define localization. What are the key components of localization? Explain.
5. Define locales. Are localization and machine translation same? Discuss.
6. Why Unicode is essential in NLP? How Unicode supports multilingual computing? Discuss encoding and its types.

Chapter 3: FSM and Morphology Introduction to NLP

7. Define Finite State Automata. Design a DFA that starts with ab and ends with bb.
8. Construct a Moore machine to generate 1's complement of a given binary number.
9. What do you mean by Morphology? Explain the types of morphological process.
10. Differentiate between Inflectional and Derivational Morphology with suitable examples.
11. Interpret the operation of below Finite-State Transducers (FST).



Chapter 4: POS and HMM

12. Why we perform Parts of Speech tagging in NLP?
13. How Part of Speech (PoS) tagging help in semantic and syntactic analysis for the below sentence?
"Can you help me with the can?".
14. Explain the different types of PoS tagging with suitable examples.

Chapter 5: Syntactic Analysis

15. What do you understand about syntactic analysis? How does Syntactic analysis work? Discuss the different processes.
16. Discuss the interaction between Lexical Analyser and Parser with a block diagram.
17. What are the main roles of parsing in NLP?
18. What do you understand by N-gram modelling? Explain unigram, bigram and Trigrams for a sentence.
19. What do you mean by one-hot encoding? Perform one-hot encoding for the sentence: "Nepal is beautiful".
20. Define parsing. Construct a parse tree for the sentence: The dog saw a man in the park.

Chapter 6: Semantic Analysis

21. What is the difference between corpus and vocabulary? Explain.
22. Discuss on hyponymy, homonymy, Synonyms and polysemy.

Chapter 7: Pragmatics and Discourse Analysis

23. Discuss about Named Entity Recognition and Word Sense Disambiguation.
24. Show your understanding about pragmatics and Discourse Analysis in NLP.
25. Explain the key concepts of pragmatics in NLP with suitable examples.
26. Briefly explain the different types of Reference in NLP.
27. Write short notes on:
 - a) Anaphoric and cataphoric reference
 - b) Challenges in reference resolution

Chapter 8: Application of NLP

28. Are opinion mining and sentiment analysis the same? How do you develop a simple sentiment analyzer? Discuss the phases.
29. Discuss the different types of sentiment analysis.
30. How aspect-based sentiment analysis contrasts from sentence-level sentiment analysis?
31. What is Text Blob? Write a simple python program to determine the polarity and subjectivity of a sentence.
32. What do you mean by polarity and subjectivity in sentiment analysis? Discuss with suitable examples.
33. Why text summarization is necessary? What are its categories?
34. Discuss a question-answering system with a block diagram.
35. Discuss some open datasets for question answering systems.
36. What is machine translation? Mention and discuss on its types.

Chapter 9: Word Vectorization and Text Preprocessing and NLTK

37. What is word vectorization? What are the different techniques for word vectorization? Discuss.
38. Text preprocessing is the initial phases of every NLP system. Justify.
39. Differentiate between supervised and unsupervised machine learning paradigms.
40. Define the terms : TF, IDF and TFIDF with suitable examples.
41. Consider the following three documents:

Document 1: "The cat sat on the mat"

Document 2: "The dog sat on the log"

Document 3: "The cat chased the dog "

Find the term frequency(TF), Document Frequency(DF), Inverse Document Frequency(IDF) and TF-IDF

42. What does cosine similarity measures? How? Explain with a suitable example.
43. What is NLTK? How do you import necessary libraries from NLTK? Illustrate some examples with python code.
44. Define tokenization. What is the difference between Word tokenizing and Sentence tokenizing.
45. Stopwords are to be removed from the text for NLP. Why?
46. What are Stems, Stemming and Stemmer? Mention any two Stemmer for English language NLP.
47. What are the fundamental differences between word stemmer and lemmatizer?
48. What are the different techniques for measuring word similarity? Discuss each in brief.
49. What do you mean by confusion matrix? How it helps in evaluating NLP model that you have built?
50. What are train and test data? Write about cross-validation technique. Compute accuracy, error rate, precision, recall and F1-score from the below confusion matrix.

		ACTUAL VALUES	
		POSITIVE	NEGATIVE
PREDICTED VALUES	POSITIVE	TP 560	FN 60
	NEGATIVE	FP 50	TN 330